ENR 1.3 INSTRUMENT FLIGHT RULES

1 Compliance with IFR

A flight conducted above flight level 200 shall be flown in compliance with IFR as prescribed in this part.

2 Aircraft Equipment

Aircraft shall be equipped with suitable instruments and radio navigation apparatus appropriate to the route to be flown and in accordance with the provisions of Regulation 91.05.1 and 91.05.2 of the Civil Aviation Regulations, 2011.

3 Change from IFR flight to VFR flight

a) The pilot-in-command of an aircraft who elects to change the conduct of flight of the aircraft from compliance with IFR to compliance with VFR shall, if a flight plan was submitted for the flight, notify the air traffic service unit concerned that the IFR flight is cancelled and communicate to such air traffic service unit the intended changes to be made to the current flight plan.

b) When an aircraft operating under IFR is flown in or encounters visual meteorological conditions, the pilot-in-command shall not cancel its IFR flight unless it is anticipated, and intended, that the flight will be continued for a reasonable period in uninterrupted visual meteorological conditions.

4 IFR procedures

a) Unless otherwise authorised by the responsible air traffic service unit, aircraft flown in compliance with the rules contained in this Division, shall comply with IFR procedures applicable in the relevant airspace.

b) Subject to the provisions of para 2, the pilot-in-command of an aircraft may execute, or endeavour to execute, a cloud break or letdown procedure at an aerodrome, or nominate an aerodrome as an alternate aerodrome: Provided that the requirements relating to cloud break or letdown procedures and to flights under IMC, as published by the Director of CAA in the NOTAM, can be complied with.

5 Air traffic service procedures

The pilot-in-command of an aircraft to be operated in controlled airspace shall -

a) ensure than an air traffic service flight plan is submitted and changes thereto are notified as prescribed in Regulation 91.03.4 of the Civil Aviation Regulations, 2011;

b) ensure that radio contact is established with the responsible air traffic service unit and that radio communication is maintained as prescribed in Regulation 91.06.16 of the Civil Aviation Regulations, 2011; and

c) comply with air traffic control clearances and instructions: Provided that-

1) the pilot-in-command of an aircraft may deviate from an air traffic control clearance in exceptional circumstances, but such deviation shall be reported to the responsible air traffic service unit as soon as possible; and
2) the pilot-in-command of an aircraft may propose an amendment to an air traffic control clearance, but such amendment shall not be applied until acceded to by the responsible air traffic service unit.

6 Mandatory radio communication in controlled airspace

6.1 The pilot-in-command of an aircraft to be operated in or crossing a controlled airspace shall ensure that, before the aircraft enters such airspace, two-way radio contact is established with the responsible air traffic service unit on the designated radio frequency, and shall ensure, while the aircraft is within, and until it leaves, the controlled airspace, that continuous radio watch is maintained and that such further two-way radio communication as such air traffic service unit may require, is established:

Provided that -

a) the air traffic service unit may permit an aircraft not capable of maintaining two-way radio communication, to fly in the control area, terminal control area, control zone or aerodrome traffic zone for which it is responsible, if traffic conditions permit, in which case the flight shall be subject to such conditions as such air traffic service unit deems necessary to ensure the safety of other air traffic; and
b) in the case of radio failure, a flight for which an air traffic service flight plan was filed and activated by the air traffic service unit on receipt of a departure time, may continue in controlled airspace if the communication failure procedures are complied with.

7 Mandatory radio communication in advisory airspace

7.1 The pilot-in-command of an aircraft to be operated in advisory airspace shall ensure that, before the aircraft approaches or enters such airspace -

a) two-way radio communication with the responsible air traffic services unit is established on the designated radio frequency;
b) if such communication is not possible, two-way radio communication is established with any air traffic service unit which is capable of relaying messages to and from the responsible air traffic unit; or

c) If such communication is not possible, broadcasts are made on the designated radio frequency giving Information on the aircrafts Intention to enter the airspace, and such pilot-in-command shall ensure that, while the aircraft is within the advisory airspace and until it departs there from, a continuous radio watch is maintained on the designated radio frequency and that-

i) such further two-way radio communication as the responsible air traffic services unit may require, is established with any other air traffic service unit which is capable of relaying messages to and from such responsible air traffic service unit;

ii) if such communication is not possible, such further two-way radio communication is established with any other air traffic service unit which is capable of relaying messages to and from the responsible air traffic service unit, as such responsible air traffic service unit may require; or

iii) if such communication is not possible, broadcasts are made on the designated radio frequency giving Information on passing reporting points and when leaving the airspace concerned: Provided that -

(aa) an aircraft maintaining a SELCAL watch while operating within an advisory route in the Johannesburg flight information region and whose SELCAL callc-sign has been communicated to the Johannesburg flight information centre, shall be deemed to be maintaining a continuous radio watch; and

(bb) in the case of a radio failure, a flight for which an air traffic service flight plan was filed and activated by an air traffic services unit on receipt of a departure time, may continue in advisory airspace if the communication failure procedures are complied with.

8 Reporting position

The pilot-in-command of an aircraft -

a) flying in controlled airspace;

b) flying in advisory airspace; or

c) on a flight for which alerting action is being provided, shall ensure that reports are made to the responsible air traffic service unit, as soon as possible, of the lime and level of passing each compulsory reporting point, together with any other information, and he or she shall further ensure that position reports are similarly made in relation to additional reporting points, if so requested by the responsible air traffic service unit and that, In the absence of designated reporting points, position reports are made at the intervals specified by the responsible air traffic service unit or published by the Director of CAA in terms of Part 175 of the Civil Aviation Regulations, 2011, for that area.

9 PERFORMANCE BASED NAVIGATION

a) In order to comply with the Navigational Specifications in South Africa, you are required to comply with the Civil Aviation Regulations and Technical Standards (CATS and CARS).

b) In the Republic of South Africa:

i) "For Oceanic procedures, RNAV 10 / RNP 4 is required to be met and complied with, ii)

ii) "For En-route procedures, RNAV5 Is reqUired to be met and compiled with,

iii) "For Terminal SID/STAR, RNAV1/RNAV2/RNP1 Is required to be met and complied with,

iv) "For Non Precision Approach procedures, RNP APCH 0.3NM Is required to be met and complied with,

v) "For Vertically Guided Approach procedures, RNP APCH with BAROVNAV I RNP AR APCH is required to be met and complied with.
Note: Special authorization to be obtained from the Director of Civil Aviation for RNP AR APCH procedures.

	Equipment Characteristics					Equipment Capability		
Class	Stand Alone	Multi Sensor	Auto Pilot REQ (Note 1)	RAIM	RAIM Equiv (Note: 2)	En-route	Terminal	Non Precision Approach
A1	X			X		X	X	X
A2	X			X		X	X	
B1		X		X		X	X	X
B2		X		X		X	X	
B3		X			X	X	X	X
B4		X			X	X	X	
C1		X	X	X		X	X	X
C2		X	X	X		X	X	
C3		X	X			X	X	X
C4		X	X			X	X	

Table 1: TSO C129/129a - Equipment Classes

Note 1: Intended to provide data to an integrated navigation system which provides enhanced guidance to an autopilot/flight director to reduce flight technical error and limited to FAR Part 121 aircraft.

Note 2: Requirement for the integrated navigation system to provide a level of GNSS integrity equivalent to RAIM

Class		Domestic Enroute	oceanic Enroute	Terminal	Departure	NPA	LNAV / NAV	Precision approach (APV - IIGLS)	ILS	Integrated sensor	Stand alone
Stand alone	Operational										
Beta	1	x	x	x	x					x	
Beta	2	x	x	x	x	x				x	
Beta	3	x	x	x	x	x	x	x		x	
Gamma	1	x	x								x
Gamma	2	x	x	x	x	x					x
Gamma	3	x	x	x	x	x	x	x			x
Delta	4								x		

Table: TSO C145/145a and C146/146a - Equipment classes

c) FAA TSO C145/145a Airborne Navigation Sensors Using the Global Positioning System (GPS) Augmented by the Wide Area Augmentation System (WAAS) This group of equipment utilises a SBAS to provide the augmentation. Functional Classes:

d) Functional Classes

Class Beta: Equipment consisting of a GPS/WAAS sensor that determines position (with integrity) and provides that position and integrity data to an integrated navigation system. In the absence of the WAAS signal integrity is provided by the use of Fault Detection and Exclusion (FDE).

Class Gamma: Equipment consisting of both the GPS/WAAS position sensor (defined by Class Beta) and a navigation function that provides path deviations relative to a selected path. Provides the navigation function of a stand-alone navigation system. In the absence of the WAAS signal integrity is provided by the use of Fault Detection and Exclusion (FDE). This class requires a database, display outputs and pilot controls.

Class Delta: Equipment consisting of both the GPS/WAAS position sensor (defined by Class Beta) and a navigation function that provides path deviations relative to a selected path. Similar to Class Gamma however, not all functions of that Class are provided. Class Delta does not provide a database or direct pilot controls. Class Delta equipment is only applicable to operational Class 4 precision approach providing an ILS replacement.

e) Operational Classes:

Class 1: Equipment that supports oceanic and domestic en route, terminal, non-precision approach and departures. This class applies the long term and fast WAAS differential corrections.

Class 2: Equipment that supports oceanic and domestic en route, terminal, non-precision approach, LNAV/VNAV, and departures. When in oceanic and domestic en route, terminal, non-precision approach and departure mode this class applies the long term and fast WAAS differential corrections. When in LNAV/VNAV this class applies the long term, fast and ionospheric corrections.

Class 3: Equipment that supports oceanic and domestic en route, terminal, non-precision approach, LNAV/VNAV, precision approach (APV-II and GLS) and departures. When in oceanic and domestic en route, terminal, non-precision approach and departure mode this class applies the long term and fast WAAS differential corrections. When in GLS, APV-II or LNAV/VNAV this class applies the long term, fast and ionospheric corrections.

Class 4: Equipment that supports precision approach operations and is intended to serve as a replacement for ILS. It is only applicable to functional Class Delta equipment.

FAA TSO C146/146a Stand-alone Airborne Navigation Equipment Using the Global Positioning System (GPS) Augmented by the Wide Area Augmentation System (WAAS)

f) These items of equipment meet the functional Class Gamma or Class Delta of FAA TSO C145/145a.

Application of navigation specification by flight phase.

Navigation Specification	Flight phase							
	En-route oceanic/ remote	En-route Continental	Arrival	Approach				
				Initial	Intermediate	Final	Missed	DEP
RNAV 10	10							
RNAV 5		5	5					
RNAV 2		2	2				1	2
RNAV 1		1	1	1		1		1
RNP 4	4							
RNP 2	2	2						
RNP 1			1	1	1		1	1
Advanced RNP	2	2 or 1	1	1	1	0.3	1	1
RNP APCH				1	1	0.3	1	
RNP AR APH				1-0.1	1-01	0.3-0.1	1-0.1	
RNP 0.3		0.3	0.3	0.3	03		0.3	0.3

9.1 Oceanic

9.1.1 RNAV-10 Specification requires

RNAV 10 was developed for operation in oceanic and remote areas and does not require any ground-based NAVAID infrastructure or assessment.

Note: As per ICAO doc 9613 RNP10 is now designated as RNAV 10.

9.1.2 RNAV 10 requires that aircraft operating in oceanic and remote areas be equipped with at least two independent and serviceable LRNSs comprising an INS, an IRS FMS or a GNSS, with integrity such that the navigation system does not provide an unacceptable probability of misleading information.

9.1.3 The assessment of a particular operator is made by the South African Civil Aviation Authority for that operator and in accordance with national operating rules (e.g. SA-CARS, JAR-OPS 1, 14 CFR Part 121) supported through appropriate advisory and guidance material. The assessment should take into account:

a) evidence of aircraft eligibility, including MEL as required by CATS
b) assessment of the operating procedures for the navigation systems to be used;
c) control of those procedures through acceptable entries in the operations manual;
d) identification of flight crew training requirements; and
e) where required, control of the navigation database process.

9.1.4 The operational approval will likely be documented through the State endorsing the air operators certificate (AOC) by issuing a Letter of Authorization, an appropriate operations specification (Ops Spec) or an amendment to the operations manual.

9.2 ENROUTE

9.2.1 RNAV 5 - Specification Requires:

VOR/DME and/or

DME/DME and/or

INS/IRS and

GNSS

Nil Navigation Database

9.2.2 The assessment of a particular operator is made by the South African Civil Aviation Authority for that operator and in accordance with national operating rules (e.g. SA-CARS, JAR-OPS 1, 14 CFR Part 121) supported through the advisory and guidance material in documents such as AMC 20-4 or AC 90-96. The assessment should take into account:

a) evidence of aircraft eligibility, including MEL as required by CATS

b) assessment of the operating procedures for the navigation systems to be used;
c) control of those procedures through acceptable entries in the operations manual;
d) identification of flight crew training requirements; and
e) when required, control of the navigation database process.

Note: As per ICAO doc 9613 B-RNAV/ RNP-5 is now designated as RNAV 5.

9.2.3 The operational approval will likely be documented through the State endorsing the air operators certificate (AOC) by issuing a Letter of Authorization, an appropriate operations specification (Ops Spec) or an amendment to the operations manual.

A navigation database does not form part of the required functionality of RNAV 5.

9.3 OCEANIC/EN ROUTE

9.3.1 RNP 4 - Specification Requires:

RNP 4 was developed for operations in oceanic and remote airspace, therefore, it does not require any ground-based NAVAID infrastructure. GNSS is the primary navigation sensor to support RNP 4, either as a stand-alone navigation system or as part of a multi-sensor system.

9.3.2 The assessment of a particular owner or operator is made by the South African Civil Aviation Authority for that owner or operator and in accordance with national operating rules (e.g. SA-CARS, JAR-OPS 1, 14 CFR Part 121) supported through appropriate advisory and guidance material. The assessment should take into account:

a) Evidence of aircraft eligibility;
b) Assessment of the operating procedures for the navigation systems to be used;
c) Control of those procedures through acceptable entries in the operations manual;
d) Identification of flight crew training requirements; and
e) Where required, control of navigation database processes;

9.3.3 The operational approval will likely be documented through the South African Civil Aviation Authority endorsing the air operator certificate (AOC) through issue of a Letter of Authorization, appropriate operations specification (Ops Spec) or amendment to the operations manual.

9.4 TERMINAL

9.4.1 RNAV 1/RNAV 2 specification requires

GNSS and/or

DME/DME and/or

DME/DME/IRU

Navigation Database

9.4.2 The RNAV 1/RNAV 2 navigation specification is primarily developed for RNAV operations in a radar environment (for SIDs, radar coverage is expected prior to the first RNAV course change). The Basic-RNP 1 navigation specification is intended for similar operations outside radar coverage

Note: For existing systems, compliance with both P-RNAV (TGL-10) and U.S. RNAV (FAA AC 90-100) assures automatic compliance with this ICAO specification.

9.4.3 The assessment of a particular operator is made by the South African Civil Aviation Authority for that operator and in accordance with national operating rules (e.g. SA-CARS, JAR-OPS 1, 14 CFR Part 121), supported through the advisory and guidance material found in documents such as TGL No. 10 and AC 90-100. The assessment should take into account:

a) evidence of aircraft eligibility, including MEL as required by CATS
b) assessment of the operating procedures for the navigation systems to be used;
c) control of those procedures through acceptable entries in the operations manual;
d) identification of flight crew training requirements; and
e) where required, control of the navigation database process

9.4.4 The operational approval will likely be documented through the State endorsing the air operators certificate (AOC) through issue of a Letter of Authorization, appropriate operations specification (Ops Spec) or amendment to the operations manual.

9.5 Basic RNP-1

9.5.1 Specification requires:

GNSS

Navigation Database

9.5.2 RNP 1 is based on GNSS positioning. Positioning data from other types of navigation sensors may be integrated with the GNSS data provided the other positioning data do not cause position errors exceeding the total system error (TSE) budget. Otherwise, means should be provided to deselect the other navigation sensor types. Operators of GNSS-equipped aircraft must have the means to predict fault detection using ABAS (e.g. RAIM)

Note: RNP 1 can be associated with Radius to Fix (RF) Path Terminator and Barometric VNAV.

RNP 1 shall not be used in areas of known navigation signal (GNSS) interference.

9.5.3 The assessment of a particular operator is made by the South African Civil Aviation Authority for that operator and in accordance with national operating rules (e.g. SA-CARS, JAR-OPS 1, 14 CFR Part 121) supported through appropriate advisory and guidance material. The assessment should take into account:

a) evidence of aircraft eligibility including MEL as required by CATS
b) assessment of the operating procedures for the navigation systems to be used;
c) control of those procedures through acceptable entries in the operations manual;
d) identification of flight crew training requirements; and
e) where required, control of the navigation database process.

9.5.4 The operational approval will likely be documented through the State endorsing the air operators certificate (AOC) through issue of a Letter of Authorization, appropriate operations specification (Ops Spec) or amendment to the operations manual.

9.5.5 The navigation database must be obtained from a supplier that complies with RTCA DO 200A/EUROCAE document ED 76, Standards for Processing Aeronautical Data, and should be compatible with the intended function of the equipment (reference ICAO Annex 6, Part 1, Chapter 7). A Letter of Acceptance (LOA), issued by the appropriate regulatory authority to each of the participants in the data chain demonstrates compliance with this requirement (e.g. FAA LOA issued in accordance with FAA AC 20-153 or EASA LOA issued in accordance with EASA IR 21 subpart G).

9.6 APPROACH

9.6.1 RNP APCH - Specification Requires:

GNSS ONLY (LNAV)

Navigation Database

9.6.2 RNP APCH is based on GNSS positioning. Positioning data from other types of navigation sensors may be integrated with the GNSS data provided the other positioning data do not cause position errors exceeding the total system error (TSE) budget, or if means are provided to deselect the other navigation sensor types.

9.6.3 The assessment of a particular operator is made by the South African Civil Aviation Authority for that operator and in accordance with national operating rules (e.g. SA-CARS, JAR-OPS 1, 14 CFR Part 121) supported through appropriate advisory and guidance material. The assessment should take into account:

a) evidence of aircraft eligibility including MEL as required by CATS;
b) assessment of the operating procedures for the navigation systems to be used;
c) control of those procedures through acceptable entries in the operations manual;
d) identification of flight crew training requirements; and
e) where required, control of the navigation database process.

9.6.4 The operational approval will likely be documented through the State endorsing the operation specifications associated with the air operator certificate (AOC) through issue of a Letter of Authorization, appropriate operations specification (Ops Spec) or amendment to the operations manual.

9.6.5 The navigation database should be obtained from a supplier that complies with RTCA DO 200A/EUROCAE document ED 76, Standards for Processing Aeronautical Data. A Letter of Acceptance (LOA) issued by the appropriate regulatory authority demonstrates compliance with this requirement (e.g. FAA LOA issued in accordance with FAA AC 20-153 or EASA LOA issued in accordance with EASA IR 21 subpart G).

9.6.6 The RNP 0.3 specification is based upon GNSS; its implementation is not dependent on the availability of SBAS. DME/DME based RNAV systems will not be capable of consistently providing RNP 0.3 performance, and States should not plan on implementing RNP 0.3 operations through application of DME/DME-based navigation. States must also not use RNP 0.3 in areas of known navigation signal (GNSS) interference. Operators relying on GNSS are required to have the means to predict the availability of GNSS fault detection (e.g. ABAS RAIM) to support operations along the RNP 0.3 ATS route. The on-board RNP system, GNSS avionics, the ANSP or other entities may provide a prediction capability. The AIP should clearly indicate when prediction capability is required and acceptable means to satisfy that requirement. This prediction will not be required where the navigation equipment can make use of SBAS augmentation and the planned operation will be contained within the service volume of the SBAS signal.

9.6.7 The following systems meet the accuracy, integrity and continuity requirements of these criteria;

a) Aircraft with E/TSO-C145a and the requirements of E/TSO-C115B FMS, installed for IFR use in accordance with FAA AC 20-130A;

b) Aircraft with E/TSO-C146a equipment installed for IFR use in accordance with FAA AC 20-138 or AC 20-138A;

c) Aircraft with RNP 0.3 capability certified or approved to equivalent standards (e.g. TSO-C193).

9.7 RNP AR APCH

9.7.1 Specification Requires:

GNSS ONLY (LNAV)

Navigation Database

9.7.2 RNP AR APCHs are only authorized based on GNSS as the primary NAVAID infrastructure. The use of DME/DME as are versionary capability may be authorized for individual operators where the infrastructure supports the required performance. RNP AR APCH shall not be used in areas of known navigation signal (GNSS) interference.Initiation of all RNP AR APCH procedures is based on GNSS updating. Except where specifically designated on a procedure as "Not Authorized", DME/DME updating can be used as a reversionary mode during the approach or missed approach when the system complies with the navigation accuracy RNP AR APCHs do not require any unique communication or ATS surveillance considerations.

9.7.3 Prior to authorization for the conduct of RNP AR APCH operations, an operator must demonstrate to the

9.7.4 South African Civil Aviation Authority that all appropriate elements of the RNP AR APCH operations have been appropriately addressed including:

a) determination of aircraft qualification;

b) training e.g. flight crews, dispatch;

c) MEL, continuing airworthiness; as required by CATS

d) requirements for operational procedures;

e) dispatch procedures;

f) maintenance procedures;

g) conditions or limitations for approval;

h) procedure operational validation for each aircraft type; and

i) conduct of a FOSA.

9.8 RNP APCH with Vertical Guidance

9.8.1 Specification Requires:

GNSS ONLY (LNAV)

BARONAV (VNAV)

Navigation Database

9.8.2 The following steps must be completed before the use of Barometric VNAV in the conduct of RNP AR APCH operations:

a) aircraft equipment eligibility must be determined and documented including MEL as required by CATS;
b) operating procedures must be documented;
c) flight crew training based upon the operating procedures must be documented;
d) the above material must be accepted by the State regulatory authority; and
e) operational approval should then be obtained in accordance with national operating rules.

9.8.3 Relevant documentation acceptable to the South African Civil Aviation Authority must be available to establish that the aircraft is equipped with an RNAV system with a demonstrated VNAV capability.

Note.- RNP AR systems: RNAV systems demonstrated and qualified for RNP AR operations including VNAV are considered qualified with recognition that the RNP approaches are expected to be performed consistent with the operators RNP AR approval. No further examination of aircraft capability, operator training, maintenance, operating procedures, databases, etc. is necessary.

9.8.4 Barometric altimetry and related equipment such as air data systems are a required basic capability and already subject to minimum equipment requirements for flight operations.

9.8.5 Following the successful completion of the above steps, an operational approval for the use of VNAV, a Letter of Authorization or appropriate operations specification (Ops Spec), or an amendment to the operations manual, if required, should then be issued by the South African Civil Aviation Authority.

10 OPERATIONAL STANDARDS FOR INERTIAL NAVIGATION AND REFERENCE SYSTEMS

10.1 General

Inertial navigation may be used by approved operators only. For approved operators of SA registered aircraft, inertial navigation may be used to satisfy the requirements of the CAA (Authority). The Inertial Navigation System (INS) or Inertial Reference System (IRS) and its installation must be certified by the state of registry as meeting Authority airworthiness standards.

Note: Airworthiness requirements of the Authority will be satisfied provided that:

The equipment has been installed to the manufacturers requirements.
The installation is listed in the aircraft Type Certificate or has a Supplemental Type Certificate for the specific aircraft type.

There is a flight manual supplement covering any system limitations.

The system is included in the operators maintenance program.

Outside SA FIRs (for example, in Europe and over the North Atlantic) other State authorities might require navigational performance different to that required by these standards.

10.2 Minimum performance for operational approval. An INS/IRS must meet the following criteria for operational approval and must be maintained to ensure performance in accordance with the criteria:

a) With a 95% probability the radial error rate is not to exceed 2NM per hour for flights up to 10 hours duration.
b) With a 95% probability the cross-track error is not to exceed +/-20 NM and along track error is not to exceed +/-25 NM at the conclusion of a flight in excess of 10 hours.

The INS/IRS should have the capability for coupling to the aircraft autopilot to provide steering guidance.

The navigation system should have the capability for updating the displayed present position.

10.3 Serviceability requirements.

An INS/IRS may be considered as serviceable for navigational purposes until such time as its radial error exceeds 3 + 3t NM (t being the hours of operation in the navigation mode).

Maintenance corrective action shall also be taken when an INS/IRS is consistently providing radial error rates in excess of 2 NM per hour and l or cross track and along track errors in excess of the tolerances given at paragraph 3.1 on more than 5% of the sectors flown.

10.4 System performance monitoring

The operator is to monitor and record the performance of INS/IRS and may be required to provide details of the system accuracies and reliabilities from time to time.

10.5 Navigation criteria

Navigation using INS/IRS as the primary navigational means is permitted in accordance with the following conditions:

a) Initial confidence Check. The INS/IRS must be checked for reasonable navigational accuracy by comparison with ground referenced radio navigation aids (which may include ATC radar) before proceeding outside the coverage of the short range radio navigation aids system.

b) Maximum Time.

1) Single INS/IRS:

a) The maximum operating time since the last ground alignment is not to exceed 10 hours.

b) On flights of more than 5 hours, any route sector may be planned for navigation by INS/IRS within the appropriate time limits (given in (c) below) but contingency navigation procedures must be available in the event of an INS/IRS in-flight unserviceability which would preclude the aircrafts operation on a subsequent route sector for which area navigation is specified.

c) INS/IRS may be used as a sole source of tracking information for continuous period not exceeding:

-3 hours in controlled airspace other than oceanic control area (OCA)

-5 hours in OCA or outside controlled airspace (OCTA)

2) Two or More INS/IRS:

a) If, during a flight, 10 hours elapsed time since the last ground alignment will be exceeded, ground alignment is to be included in the pre-flight cockpit procedures prior to push back taxi for departure.

b) INS/IRS may be used as the sole source of tracking information for continuous periods not exceeding:

-5 hours in controlled airspace other than OCA.

-12 hours in OCA or OCTA.

Notes:

1) **Provided that the use of INS/IRS as the sole means of navigation does not exceed the time limit, the aircraft may be operated for longer periods using the INS/IRS with either manual or automatic updating.**
2) **The 5 hour limit on single INS/IRS ensures 99.74% (3 sigma) probability that loss of satisfactory navigational capability will not occur with equipment mean time between failures (MTBF) of approximately 1900 hours. If the demonstrated MTBF exceeds 2000 hours, the maximum time may be increased.**

c) Updating Present Position. Updating inertial present position in flight is permitted in the following instances only:

1) Manually:

a) **Overhead a VOR beacon.**
b) **Within 25 nautical miles of a co-located VOR/DME beacon.**
c) **Over a visual fix when at a height not more that 5000 ft above the feature.**

2) Automatically:

a) **Within 200 nautical miles of a DME site when the aircraft strack will pass within 140 NM of the site.**
b) **Within 200 NM of both DME sites for a DME/DME fix.**
c) **From a co-located VOR/DME beacon provided that updates from a receding beacon are not accepted when the beacon is more that 25 NM from the aircraft.**

Notes:

1) **En Route VOR and DME sites separated by not more that 500 metres are considered to be collocated.**
2) **DME slant range error correction might be necessary in some circumstances.**
3) **Updating a present position from a visual fix may not be planned for IFR flights.**
4) **A receding beacon is one from which the distance to the aircraft is increasing.**
5) **Updating in other circumstances (for example, over a NDB) will not provide sufficient accuracy to ensure that the INS/IRS operates within the prescribed tolerances for navigation.**
6) **Because INS/IRS are essentially accurate and reliable, and ground alignment is more accurate than in-flight updating, updating of present position is usually not warranted especially during the initial few hours of operation. However, INS/IRS errors generally increase with time and are not self-**

correcting. Unless the error is fairly significant (for example, more than 4 NM or 2 NM/hr) it may be preferable to retain the error rather than manually update.

d) Limitation on Use. Wherever track guidance is provided by radio navigation aids, the pilot-in-command shall ensure that the aircraft remains within the appropriate track-keeping tolerances of the radio navigation aids. INS/IRS is not to be used as a primary navigation reference during IFR flight below lowest safe altitude (LSALT).

e) Pre-flight and En-route Procedures.

The following practices are required:

1) New data entries are to be cross-checked between at least two crew members for accuracy and reasonableness, or, for single pilot operations, an independent check (for example, of INS/IRS- computed tracks and distances against the flight plan) must be made.

2) As a minimum, position and tracking information is to be checked for reasonableness (confidence check) in the following cases:

a) Prior to each compulsory reporting point.

b) At or prior to arrival at each en-route way point during RNAV operation along RNAV routes.

c) At hourly intervals during area type operation off established RNAV routes.

d) After insertion of new data.

10.6 OPERATING CRITERIA

10.6.1 Two or more INS / IRS Installations. For two or more INS / IRS installations:

a) If one INS/IRS fails or can be determined to have exceeded a radial error of 3+3t NM, operations may continue on area navigation routes using the serviceable system(s) in accordance with the navigational criteria applicable to the number of INS/IRS units remaining serviceable.

b) If,

1) The difference of pure inertial read-outs between each pair of INS/IRS is less than 1.4 (3+3t) NM, no action is required.

2) The difference of pure inertial read-outs between any pair of INS/IRS exceeds 1.4 (3+3t) NM and it is possible to confirm that one INS/IRS has an excessive drift error, that system should be disregarded and/or isolated from the other system(s) and the apparently serviceable system(s) should be used for navigation.

Note: 1) This check and its isolation action are unnecessary if a multiple INS/IRS installation is protected by a serviceability self test algorithm.

3) Neither condition (1) nor (2) can be satisfied, another means of navigation should be used, and the pilot-in-command must advise the appropriate ATS unit.

10.6.2 Single INS/IRS installations

For single INS/IRS installations, if the INS/IRS fails or exceeds the serviceability tolerance:

a) The pilot-in-command must advise the appropriate ATS unit of INS/IRS failure.

b) Another means of navigation is to be used.

c) The aircraft is not to begin a route sector for which area navigation is specified unless it is equipped with an alternative, serviceable, approved area navigation system.

10.6.3 Autopilot Coupling: Autopilot coupling to the INS/IRS should be used, whenever practicable, if this feature is available. If for any reason the aircraft is flown without autopilot coupling, the aircraft is to be flown within an indicated cross-track tolerance of +/-2 NM. In controlled airspace ATC is to be advised if this tolerance is exceeded.

10.7 NAVIGATIONAL TOLERANCES

10.7.1 The maximum drift rate expected from INS/IRS is 2 NM per hour (2 sigma probability). For the purposes of navigation and determining aircraft separation, the 3 sigma figure of 3 NM is allowed so that the maximum radial error with 3 sigma confidence equals 3+3t NM where t equals the time in hours since the INS/IRS was switched into the navigation mode.

10.7.2 DME and other inputs can automatically influence the INS/IRS to improve the accuracy of its computed position. The pilot may also insert known position co-ordinates to update the INS/IRS. Therefore, if the system is updated with known position information the position error is reduced and the INS/IRS can be assumed to operate within the radial error tolerance of 3+3T NM where T is the time (hours elapsed since the last position update.)

10.7.3 The accuracy of the data used for updating must be considered. The navigation aid positions used for updating inertial present position are accurate to within 0.1 NM. However, the aircraft in flight cannot be fixed to the same order of magnitude. The accuracy of the position fix is taken as +/- 3 NM radial error.

10.7.4 Because the INS/IRS error, the navigation aid position accuracy and the position fix errors are independent of each other, the total radial error is determined by the root-sum-square method:

$$\text{Total error} = \sqrt{(3 + 3T)^2 + 0.1^2 + 3^2}\ \text{nm}$$

10.7.5 The effect of navigation aid position accuracy on the total error is negligible, and so,

$$\text{totalerror} = \sqrt{(3 + 3T)^2 + 3^2}\ \text{nm}$$
$$= \sqrt{(1 + T)^2 + 1}\ \text{nm}$$

Substituting values for T

at time of update, total radial error = 4.2 NM

after 1 hour = 6. 7 NM

after 2 hours = 9.5 NM

after 3 hours = 12.4 NM

after 4 hours = 15.3 NM

after 5 hours = 18.2 NM

after 6 hours = 21.2 NM

10.7.6 Dual installation: If two INS/IRS are installed and the aircraft is navigated by averaging, the inertial present position formula for the total radial error given in paragraph 7.4 is modified by multiplying by:

$$\frac{1}{\sqrt{}} (=0.7)$$

10.7.7 Triple installations: If three INS/IRS are installed and triple mix is used, the total radial error is further reduced. For simplicity for navigation and aircraft separation the tolerances applicable to dual installations apply and the third system provides redundancy.

10.8 GLOBAL NAVIGATION SATELLITE SYSTEM (GNSS) AND THE USE OF GLOBAL POSITIONING SYSTEM (GPS)

10.8.1 When the airborne navigation equipment using GPS is CAA (South African Civil Aviation Authority) approved as satisfying the relevant technical criteria, then operators may be approved to conduct flights when flying under IFR in oceanic, domestic en-route airspace and under certain circumstances in terminal and approach airspace as per the following paragraph Operational Matters.

The criteria presently specified may be superseded by Airworthiness and Operational Standards promulgated by the CAA.

10.8.2 AIRWORTHINESS APPROVAL

FAA Advisory Circulars AC 20-138/AC 20-138A (GPS stand-alone system) or AC 20-130A (Multi-sensors systems) is used as the basis for the airworthiness approval of an RNAV system based on GNSS. For APV BARO-VNAV operation, FAA Advisory Circular AC 20-129 is used as the airworthiness basis with additional requirements.

a) Stand alone Equipment:

1) If the RNAV installation is based on GNSS stand-alone system, the equipment shall be approved in accordance with TSO-C129a/ETSO-C129a or ETSO-C146/TSO-C146 as per the below table.

b) Multi-sensor Equipment:

2) If the RNAV installation is based on GNSS sensor equipment used in a multi-sensor system (e.g. FMS), the GNSS sensor shall be approved in accordance with TSO-C129/ETSO-C129 or ETSO-C145/TSO-C145 as per the table below.

3) Multi-sensor systems using GNSS should be approved in accordance with AC20-130A or ETSO-C115b/TSO-C115b, as well as having been demonstrated for RNP capability.

Note 1: For GNSS receiver approved in accordance with ETSO-C129/TSO-C129, the capability for satellite Fault Detection and Exclusion (FDE) is recommended, to improve Continuity of function.

Note 2: GNSS receivers approved in accordance with ETSO-145/TSO-C145a or ETSO- C146/TSOC146a (DO 229C) and used outside SBAS coverage area may trigger inappropriate Loss of Integrity (LOI) warnings. DO229D paragraph 2.1.1.6 provides a correct satellite selection scheme requirement to address this issue. Although most of the ETSO-C145/TSO-C145a or ETSO-146/TSO-C146a approved receivers comply with this satellite selection scheme, a confirmatory statement from the equipment manufacturer is still necessary. It should be noted that such confirmatory statement is not necessary for equipment compliant with TSO-C145b or TSO-C146b.

10.9 REQUIREMENTS FOR USE OF GLOBAL POSITIONING SYSTEM (GPS) AS AN APPROVED PRIMARY MEANS IFR NAVIGATION AID

10.9.1 PURPOSE

The purpose of this section is to detail the Civil Aviation Authorities requirements for the use of GPS as an approved en-route and area IFR primary means navigation aid.

10.9.2 This constitutes the Civil Aviation Authority (CAA) approval for the use of a GPS system, fitted and operated in accordance with the provisions of this within South Africa domestic airspace only, for the purpose of:

a) position fixing, as required in AIP GEN 1.5 & ENR 1.3;

b) long range navigation including operations on designated RNP routes;

c) deriving distance information, for en route navigation, traffic information and ATC separation;

d) application of RNP based separation.

10.9.3 BACKGROUND

GPS has been confirmed for IFR en route supplemental navigation use in South Africa. As further information has become available on the accuracy, integrity, availability and continuity of GPS, and following US DoD declaration of Full Operational Capability (FOG) in April 1995, CAA has determined that the use of GPS for IFR navigation can be extended, in accordance with the provisions of this section.

Instrument arrival procedures using GPS derived distance combined with NDB or VOR azimuth information, will be introduced at some future date, and will be known as GPS Arrival procedures.

10.9.4 GPS SIGNAL INTEGRITY

System integrity is an essential element of the approval for use of GPS as a primary means navigation system. GPS receivers certified to TSO-C129 provide integrity through the use of RAIM, or an approved equivalent integrity system. When RAIM is lost or not available, the accuracy of the system cannot be assumed to meet the required standard for navigation, or for the application of ATC separation standards.

GPS integrity is also dependent on the number of operational satellites in view, or available for use. Loss of one or more satellites can result in degraded system availability.

RAIM availability is greatly improved through the use of barometric aiding. Except as provided in this section, GPS must not be used to fix position, provide distance information or provide primary navigation, unless RAIM is available.

10.9.5 GPS SATELLITE CONSTELLATION

The approvals contained in this section are based on the availability of the US DoD GPS Standard Positioning Service (SPS) operating to its defined Full Operational Capability (FOG). This service does not meet the requirements of a sole means navigation system.

Disruption to the GPS may result in degradation in GPS service to such a level that some or all of the operational approvals for the IFR primary use of GPS contained in the section may need to be withdrawn. When known, these changes or restrictions will be advised by NOTAM.

Prior knowledge of RAIM availability will enable operators to use the system more efficiently, by allowing operations to be planned around gaps in RAIM coverage (RAIM holes). To achieve these efficiencies, CAA recommends that appropriate RAIM prediction capabilities be available at dispatch locations. Flights should be planned to ensure the safe completion of flight in the event of Joss of GPS integrity.

10.10 PILOT TRAINING

The following pilot training requirements must be satisfied:

a) Prior to using GPS in IFR operations for any of the purposes specified in this section, the holder of an instrument rating must, unless exempted by CAA, have completed a course of ground training based on the syllabus contained in Annex A. The course must be conducted by, or on behalf of, an approved IFR check and training organisation or approved instrument training school or by the CAA.

b) The course must cover both general information and procedures applicable to all types of GPS equipment, as well as the essential operating procedures for a specific type of aircraft equipment. Pilots who have completed the course and who wish to use a different type of GPS aircraft equipment must

ensure that they are familiar with, and competent in, the operating procedures required for that type of equipment, before using it in flight for any of the purposes approved in this section.

c) Licensing or certification requirements for pilots will be issued in due course by CAA.

10.11 OPERATIONAL REQUIREMENTS

The following operational requirements must be satisfied:

a) Operating instructions for GPS navigation equipment must be:

1) carried on board; and

2) incorporated into the Company Operations Manual for commercial operations.

b) GPS navigation equipment must be operated in accordance with the operating instructions, and any additional requirements specified in the approved aircraft flight manual or flight manual supplement.

c) In addition to GPS, aircraft must be equipped with serviceable radio navigation systems as specified by the authority.

d) When within rated coverage of ground based navigation aids, pilots must monitor the ground based system, and maintain track as defined by the most accurate ground based radio navigation aid (VOR or NDB) available. If there is a discrepancy between the GPS and ground based system information, pilots must use the information provided by the ground based navigation system.

e) ATC may require GPS equipped aircraft to establish on, and track with reference to, a particular VOR radial or NDB track for the application of separation.

10.12 OPERATIONS WITHOUT RAIM

10.12.1

a) Navigation (Nav) Solution with RAIM;

b) 2D or 3D Nav Solution without RAIM; and

c) Dead Reckoning (DR), or Loss of Nav Solution.

10.12.2 ATS services, and in particular ATC separation standards, are dependent on accurate navigation and position fixing. If RAIM is lost, the accuracy of the system is assumed not to meet the required standard for both navigation and application of ATC separation. Accordingly, when RAIM is lost, the following procedures must be adopted:

a) Aircraft tracking must be closely monitored against other on-board systems.

b) In controlled airspace, ATC must be advised if:

i) RAIM is lost for periods greater than ten minutes, even if GPS is still providing positional information; or

ii) RAIM Is not available when ATC request GPS distance, or if an ATC clearance or requirement based on GPS distance is imposed; or

iii) the GPS receiver is in DR mode, or experiences loss of navigation function, for more than one minute; or

iv) Indicated displacement from track centre line is found to exceed 2NM.

10.12.3 ATC may then adjust separation:

a) If valid position information is lost (2D and DR Mode), or non RAIM operation exceeds ten minutes, the GPS information is to be considered unreliable, and another means of navigation should be used until RAIM is restored and the aircraft is re-established on track.

b) Following re-establishment of RAIM, the appropriate ATS unit should be notified of RAIM restoration, prior to using GPS Information. This will allow ATC to reassess the appropriate separation standards.

c) When advising ATS of the status of GPS the phrases RAIM FAILURE or RAIM RESTORED must be used.

10.13 GPS DISTANCE INFORMATION TO AIS UNITS

10.13.1 When a DME distance is requested by an ATS unit, DME derived distance information should normally be provided. Alternatively, GPS derived distance information may be provided to an ATS unit, unless RAIM is currently unavailable and has been unavailable for the preceding ten minutes.

10.13.2Notwithstanding para 10.13.1, if an ATC unit has issued a clearance or requirement based upon GPS distance (e.g. a requirement to reach a certain level by a GPS distance), pilots must inform ATC if RAIM is not available.)

10.13.3 When a DME distance is not specifically requested, or when the provision of a DME distance is not possible, distance information based on GPS derived information may be provided. When providing GPS distance, transmission of distance information must include the source and point of reference -(e.g. 115 NM GPS JSV, 80 NM GPS VAL NDB, 267 NM GPS ORNAD e.t.c.)

10.13.4 If a GPS distance is provided to an ATC unit, and RAIM Is not currently available, but has been available in the preceding 10 minutes, the distance report should be suffixed NEGATIVE RAIM- (e.g. 26 NM GPS BLV NEGATIVE RAIM.)

10.13.5 Databases sometimes contain waypoint information which is not shown on published AIP charts and maps. Distance information must only be provided in relation to published waypoints unless specifically requested by an ATS unit.

10.13.6 Where GPS distance is requested or provided from an NDB, VOR, DME, or published waypoint, the latitude and longitude of the navigation aid or waypoint must be derived from a validated database which cannot be modified by the operator or crew (refer para 11.7 below).

10.14 DATA INTEGRITY

10.14.1 As a significant number of data errors, in general applications, occur as a result of manual data entry errors, navigation aid and waypoint latitude and longitude data should be derived from a data base, if available, which cannot be modified by the operator or crew.

10.14.2 When data is entered manually, data entries must be crosschecked by at least two crew members for accuracy and reasonableness, or, for single pilot operations; an independent check (e.g. GPS computed tracks and distances against current chart data) must be made.

10.14.3 Both manually entered and database derived position and tracking information should be checked for reasonableness (confidence check) in the following cases:

a) prior to each compulsory reporting point;
b) at or prior to arrival at each en route waypoint;
c) at hourly intervals during area type operations when operating off established routes; and
d) after insertion of new data - (e.g. creation of new flight plan.)

10.15 FLIGHT PLAN NOTIFICATION

10.15.1 Flight Plan notification will be in compliance with ICAO Doc 4444 - Flight Planning and South African Aeronautical Information Circular AIC D003/2022

10.16 REPORTING

10.16.1 Approved users are invited to submit details of any anomalies experienced during the use of GPS and/or any other comments in writing to the CAA for evaluation to the following address:

Civil Aviation Authority

Private Bag x 73

Halfway House

1685

Fax +2711 5451459 for attention: Director of Civil Aviation

10.17 ANNEX A - Syllabus of training GPS as Primary Means Navigation

10.17.1 GPS System Components and Principle of Operation.Demonstrate an understanding of the GPS system and its principles of operation:

a) GPS system components, constellation, control and user.
b) Aircraft equipment requirements.
c) GPS satellite signal and pseudo random code.
d) Principle of position fixing.
e) Method of minimising receiver clock error.
f) Minimum satellites required for navigation functions.
g) Masking function.
h) Performance limitations of various equipment types.
i) GPS use of WGS84 co-ordinate system.

10.17.2 Navigation System Performance Requirements.Define the following terms in relation to a navigational system, and recall to what extent the GPS system meets the associated requirements:

a) Accuracy.
b) Integrity. - Means of providing GPS integrity; RAIM; procedural systems integration.
c) Availability.
d) Continuity of service.

10.17.3 Authorisation and Documentation

Recall the requirements applicable to pilots and equipment for GPS OPS:

a) Pilot training requirements
b) Log book certification
c) Aircraft equipment requirements
d) GPS NOTAM.

10.17.4 GPS Errors and Limitations

Recall the cause and magnitude of typical GPS errors:

a) Ephemeris
b) Clock
c) Receiver
d) Atmospheric lionospheric
e) Multipath
f) SA (Selected Availability)
g) Typical Total error associated with CIA code
h) Effect of PDOP I GDOP on position accuracy
i) Susceptibility to interference
j) Comparison of vertical and horizontal errors
k) Tracking accuracy and collision avoidance.

10.17.5 Human Factors and GPS

Be aware of the human factors limitations associated with the use of GPS equipment. Apply GPS operating procedures which provide safeguards against navigational errors and loss of situational awareness due to these causes:

a) Mode errors
b) Data entry errors
c) Data validation and checking including independent cross checking procedures
d) Automation induced complacency
e) Non-standardisation of the GPS o pilot interface
f) Human information processing and situational awareness.

10.17.6 GPS Equipment Specific Navigation Procedures

Recall and apply knowledge of appropriate GPS operating procedures to typical navigational tasks using a specific type of aircraft equipment by:

a) Select appropriate operational modes
b) Recall categories of information contained in the navigational database
c) Predict RAIM availability
d) Enter and check user defined waypoints
e) Enter, retrieve and check flight plan data
f) Interpret typical GPS navigational displays LAT LONG, distance and bearing to waypoint, CDI
g) Intercept and maintain GPS defined tracks
h) Determine TMG, GS, ETA, time and distance to WPT, WV inflight
i) Indications of waypoint passage
j) Use of direct to function

k) Use of nearest airport function
l) Use of GPS in GPS and VOR / DME / GPS arrival procedures.

10.17.7 GPS Equipment Checks

For the specific type of aircraft equipment, carry out the following GPS operational and serviceability checks at appropriate times:

a) TSO status
b) Satellites acquired
c) RAIM status
d) PDOP I GDOP status
e) IFR Database currency
f) Receiver serviceability
g) CDI sensitivity
h) Position indication

10.17.8 GPS Warning and Messages

For the specific type of aircraft equipment, recognise and take appropriate action for GPS warnings and messages, including the following:

a) Loss of RAIM
b) 2D navigation
c) In Dead Reckoning mode
d) Database out of date
e) Database missing
f) GPS fail
g) Barometric input fail
h) Power I battery fail
i) Parallel offset on
j) Satellite fail

10.18 GLOSSARY

Active Waypoint -The waypoint to/from which the navigational guidance is being provided.

Along Track Distance (ATD) Fix - A distance in nautical miles (NM) to the active waypoint along the specified track. An ATD fix will not be used where a course change is made.

Course Set - Guidance set from information provided by the GPS equipment that assists the pilot in navigating to or from an active waypoint on a heading/bearing.

Dead Reckoning (DR) - The navigation of an aircraft solely by means of computations based on airspeed, course, heading, wind direction and speed, ground speed and elapsed time.

Direct To-A method used with the GPS equipment to provide the necessary course from present position directly to a selected waypoint. This is not the course waypoint to waypoint.

En Route Domestic - The phase of flight between departure and arrival terminal phases, with departure and arrival points within the RSA Airspace.

En Route Oceanic - The phase of flight between the departure and arrival terminal phases with an extended flight route over the high seas.

En Route Operations- The phase of navigation covering operations between departure and arrival terminal phases. The en route phase of navigation has two subcategories: en route domestic and en route oceanic.

Fly By Waypoint - A waypoint which requires turn anticipation to allow tangential interception of the next segment of a route or procedure. or.

Fly Over Waypoint - A waypoint at which a turn is initiated in order to join the next segment of a route or procedure.

Geodetic Datum - A minimum set of parameters required to define location and orientation of the local reference system with respect to the global reference system/frame.

Global Navigation Satellite Systems (GNSS) -An umbrella term adopted by the International Civil Aviation Organization (ICAO) to encompass any independent satellite navigation system used by a pilot to perform on board position determinations from the satellite data.

Global Positioning system (GPS)- a U.S. space-based positioning, velocity and time system composed of space, control, and user elements. The space element, when fully operational will be composed of 24 satellites in six orbital planes. The control element

consists of five monitor stations, three ground antennas and a master control station. The user element consists of antennas and receiver-processors that provide positioning, velocity, and precise timing to the user.

Integrity- The probability that the system will provide accurate navigation as specified or timely warnings to users when GPS data should not be used for navigation.

Minimum en-route altitude (MEA)- The altitude for an en-route segment that provides adequate reception of relevant navigation facilities and ATS communications complies with the airspace structure and provides the required obstacle clearance.

Minimum obstacle clearance altitude (MOCA)- The minimum altitude for a defined segment of flight that provides the required obstacle clearance.

Non-precision Approach Operations - Those flight phases conducted on charted Standard Instrument Approach Procedures (SIAPs) commencing at the initial approach fix and concluding at the missed approach point or the missed approach holding point, as appropriated.

Oceanic Airspace - Airspace over the oceans of the world, considered international airspace, where oceanic separation and procedures per the International Civil Aviation Organization (ICAO) are applied. Responsibility for the provisions of air traffic control service in this airspace is delegated to various countries, based generally upon geographic proximity and the availability of the required resources.

Receiver Autonomous Integrity Monitoring (RAIM)-A technique whereby a civil GPS receiver/processor determines the integrity of the GPS navigation signals using only GPS signals or GPS signals augmented with altitude. This determination is achieved by a consistency check among redundant pseudo range measurements. At least one satellite in addition to those required for navigation must be in view for the receiver to perform the RAIM function.

Selective Availability (SA)- A method by which the U.S. Department of Defence can artificially create a significant time and positioning error in the satellites. This feature is designed to deny an enemy the use of precise GPS positioning data.

Supplemental Air Navigation System- An approved navigation system that can be used in conjunction with, or in addition to a primary air navigation system.

TO - FROM Navigation- RNAV equipment in which the desired path over the ground in defined as a specific (input quantity) course emanating either to or from a particular waypoint. The equipment functions like a conventional VOR receiver whore the CDI needle and the "to/ from" indicator responds to movement of the OBS. In this equipment the aircraft may fly either TO or FROM any single designated waypoint.

TO-TO Navigation- RNAV equipment in which a path is computed that connects two waypoints. In this equipment, two waypoints must always be available, and the aircraft is usually flying between the two waypoints and to the active waypoint. In this equipment the CDI needle functions like its tracking a localizer signal; that is movement of the OBS has no effect on the CDI needle or the "to/from".

Turn Anticipation - The capability of RNAV systems to determine the point along a course, prior to a turn waypoint, whore a turn should be initiated to provide a smooth path to intercept the succeeding course within the protected airspace.and to enunciate the information to the pilot.

User-selectable Navigation Database- A navigation database having user defined contents accessible by the pilot and/or the navigation computer during aircraft operations in support of navigation needs. This database is stored electronically and is typically updated at regular intervals, such as the AIRAC 28 day cycle. It does not include data that can be entered manually by the pilot or operator.

Waypoint (wp)- A Specified geographical location used to define an area navigation route or the flight path of an aircraft employing area navigation.Waypoints are identified as either Fly-by waypoint or fly-over waypoint.

World Geodetic System (WGS)-A consistent set of parameters describing the size and shape of the earth, the positions of a network of points with respect to the centre of mass of the earth, transformations from major geodetic datum, and the potential of the earth (usually in terms of harmonic coefficients).